MINAL AMJAD

FA22-BCS-037

QUESTION 1:

ANSWER:

In [2]:
```
import numpy as np

# initial guesses
x1 = 0
x2 = 0
x3 = 0

epsilon = 0.01
converged = False

x_old = np.array([x1, x2, x3])

print('Iteration results')
print('k,    x1,    x2,    x3')

for k in range(1, 50):

    x1 = (14 - 3*x2 + 3*x3) / 8
    x2 = (5 + 2*x1 - 5*x3) / (-8)
    x3 = (-8 - 3*x1 - 5*x2) / 10

    x = np.array([x1, x2, x3])

    # error calculation
    dx = np.sqrt(np.dot(x - x_old, x - x_old))
```

```
    print("%d, %.4f, %.4f, %.4f" % (k, x1, x2, x3))

    if dx < epsilon:
        converged = True
        print("Converged!")
        break

    x_old = x

if not converged:
    print("Not converged, increase the number of iterations")
```

```
Iteration results
k,    x1,     x2,     x3
1, 1.7500, -1.0625, -0.7937
2, 1.8508, -1.5838, -0.5633
3, 2.1327, -1.5103, -0.6847
4, 2.0596, -1.5678, -0.6340
5, 2.1002, -1.5463, -0.6569
6, 2.0835, -1.5565, -0.6468
7, 2.0911, -1.5520, -0.6513
Converged!
```

QUESTION 2:

ANSWER:

In [3]:

```
import numpy as np

a = [[8, 3, -3], [-2, -8, 5], [3, 5, 10]]

# Find diagonal coefficients
diag = np.diag(np.abs(a))

# Find row sum without diagonal
off_diag = np.sum(np.abs(a), axis=1) - diag

if np.all(diag > off_diag):
    print('matrix is diagonally dominant')
```

```
else:
    print('NOT diagonally dominant')
```

```
matrix is diagonally dominant
```

QUESTION 11.9

Solve the following system of equations using the Gauss-Seidel Method.

Continue the iterations until the approximate relative error is less than 5%.

$$15c_1 - 3c_2 - c_3 = 3800$$

$$-3c_1 + 18c_2 - 6c_3 = 1200$$

$$-4c_1 - c_2 + 12c_3 = 2350$$

ANSWER:

In [4]:

```
import numpy as np

c1 = 0
c2 = 0
c3 = 0

epsilon = 0.05
converged = False

x_old = np.array([c1,c2,c3])

print('Iteration results')
print('k, c1, c2, c3')

for k in range(1,50):

    c1 = (3800 + 3*c2 + c3)/15
    c2 = (1200 + 3*c1 + 6*c3)/18
    c3 = (2350 + 4*c1 + c2)/12
```

```
    x = np.array([c1,c2,c3])

    dx = np.sqrt(np.dot(x-x_old,x-x_old))

    print("%d, %.4f, %.4f, %.4f"%(k,c1,c2,c3))

    if dx < epsilon:
        converged = True
        print("Converged!")
        break

    x_old = x

if not converged:
    print("Not converged")
```

```
Iteration results
k, c1, c2, c3
1, 253.3333, 108.8889, 289.3519
2, 294.4012, 212.1842, 311.6491
3, 316.5468, 223.3075, 319.9579
4, 319.3254, 226.5402, 321.1535
5, 320.0516, 227.0598, 321.4388
6, 320.1745, 227.1754, 321.4895
7, 320.2010, 227.1967, 321.5001
Converged!
```

QUESTION 11.11

Solve the following system of linear equations using the Gauss_Seidel Method.

Continue the iterations until the approximate relative error is less than 5%.

$$10x_1 + 2x_2 - x_3 = 27$$

$$-3x_1 - 6x_2 + 2x_3 = -61.5$$

$$x_1 + x_2 + 5x_3 = -21.5$$

ANSWER:

In [5]:
```
import numpy as np

x1 = 0
x2 = 0
x3 = 0

epsilon = 0.05
converged = False

x_old = np.array([x1,x2,x3])

print('Iteration results')
print('k, x1, x2, x3')

for k in range(1,50):

    x1 = (27 -2*x2 + x3)/10
    x2 = (61.5 -3*x1 +2*x3)/6
    x3 = (-21.5 -x1 -x2)/5

    x = np.array([x1,x2,x3])

    dx = np.sqrt(np.dot(x-x_old,x-x_old))

    print("%d, %.4f, %.4f, %.4f"%(k,x1,x2,x3))

    if dx < epsilon:
        converged = True
        print("Converged!")
        break

    x_old = x

if not converged:
    print("Not converged")
```

```
Iteration results
k, x1, x2, x3
1, 2.7000, 8.9000, -6.6200
2, 0.2580, 7.9143, -5.9345
3, 0.5237, 8.0100, -6.0067
4, 0.4973, 7.9991, -5.9993
Converged!
```

QUESTION 11.12

Solve the following system of equations using the Gauss-Seidel Method with relaxation.

Use a relaxation parameter

$$\lambda = 0.95$$

Continue the iterations until the approximate relative error is less than 5%.

$$10x_1 + 2x_2 - x_3 = 27$$

$$-3x_1 - 6x_2 + 2x_3 = -61.5$$

$$x_1 + x_2 + 5x_3 = -21.5$$

ANSWER:

In [6]:
```python
import numpy as np

x1 = 0
x2 = 0
x3 = 0

lam = 0.95

epsilon = 0.05
converged = False

x_old = np.array([x1,x2,x3])
```

```python
print('Iteration results')
print('k, x1, x2, x3')

for k in range(1,50):

    x1_gs = (27 -2*x2 + x3)/10
    x1 = lam*x1_gs + (1-lam)*x1

    x2_gs = (61.5 -3*x1 +2*x3)/6
    x2 = lam*x2_gs + (1-lam)*x2

    x3_gs = (-21.5 -x1 -x2)/5
    x3 = lam*x3_gs + (1-lam)*x3

    x = np.array([x1,x2,x3])

    dx = np.sqrt(np.dot(x-x_old,x-x_old))

    print("%d, %.4f, %.4f, %.4f"%(k,x1,x2,x3))

    if dx < epsilon:
        converged = True
        print("Converged!")
        break

    x_old = x

if not converged:
    print("Not converged")
```

```
Iteration results
k, x1, x2, x3
1, 2.5650, 8.5191, -6.1910
2, 0.4865, 7.9719, -6.0016
3, 0.5045, 7.9959, -6.0002
Converged!
```

QUESTION 11.13

Investigate whether the following systems of equations are diagonally dominant.

Determine whether the Gauss-Seidel method will converge.

Example system

$$8x_1 - 3x_2 + 2x_3 = 12$$

$$6x_1 + 7x_2 - x_3 = -1$$

$$2x_1 + 4x_2 - x_3 = 5$$

ANSWER:

In [7]:

```
import numpy as np

x1 = 0
x2 = 0
x3 = 0

epsilon = 0.05
converged = False

x_old = np.array([x1,x2,x3])

print('Iteration results')
print('k, x1, x2, x3')

for k in range(1,50):

    x1 = (12 +3*x2 -2*x3)/8
    x2 = (-1 -6*x1 +x3)/7
    x3 = 2*x1 +4*x2 -5

    x = np.array([x1,x2,x3])

    dx = np.sqrt(np.dot(x-x_old,x-x_old))

    print("%d, %.4f, %.4f, %.4f"%(k,x1,x2,x3))

    if dx < epsilon:
```

```
        converged = True
        print("Converged!")
        break

    x_old = x

if not converged:
    print("Not converged")
```

```
Iteration results
k, x1, x2, x3
1, 1.5000, -1.4286, -7.7143
2, 2.8929, -3.7245, -14.1122
3, 3.6314, -5.2715, -18.8233
4, 4.2290, -6.4567, -22.3690
5, 4.6710, -7.3421, -25.0265
6, 5.0033, -8.0067, -27.0199
7, 5.2525, -8.5050, -28.5149
8, 5.4394, -8.8787, -29.6362
9, 5.5795, -9.1591, -30.4772
10, 5.6846, -9.3693, -31.1079
11, 5.7635, -9.5270, -31.5809
12, 5.8226, -9.6452, -31.9357
13, 5.8670, -9.7339, -32.2018
14, 5.9002, -9.8004, -32.4013
15, 5.9252, -9.8503, -32.5510
16, 5.9439, -9.8877, -32.6632
17, 5.9579, -9.9158, -32.7474
18, 5.9684, -9.9369, -32.8106
19, 5.9763, -9.9526, -32.8579
20, 5.9822, -9.9645, -32.8934
Converged!
```

In [ ]: